

# Charles checklist — post-PD16 (manual checkoff)

Last synced: 2026-08-04

Source: Paper Directions 16 (Aug 4, 2026) — `../transcripts/PD16_notes.md`; narrative `08_PD16_Alex_meeting_takeaways.md`

Supersedes: `archive/checklists/CHARLES_CHECKLIST_archive_20260804_pd15.md` (PD15 characterization deck arc).

Where we are: Phase B characterization deck is built (intro +  $\rho$ ,  $\lambda$ ,  $\theta$ ,  $\theta \times K/N$ ,  $\gamma$ ,  $\lambda_{crit}$ ) and you briefed Alex (PD16). Alex accepted the deck structure and sort-and-chop story. Near-term mission shifted: implement team-level L\_C,  $\theta$  from K/N, and L\_C-vs- $\rho$  distribution diagnostics — then move toward data-linked  $\rho$ ,  $\theta$  and principled  $\gamma$ ,  $\lambda$  (not “match the hero curve” as the definition).

How to use

1. Work top to bottom.
2. Change `[ ]` → `[x]` only when you finished that row.
3. Fill Proof with a path, date, or one-line note.
4. Two decks: `So_Far_.pptx` (Pass A/B/C gallery, slides 1–3) ≠ `CHAR_Phase_B_characterization.pptx` (OAT sensitivity deck).
5. Score:  $S_i = A_i - \lambda L_C$ ;  $\lambda$  = weight on congestion; L\_C = congestion (smooth sigmoid in current deck).

Notation: `02_Three_Kinds_of_Model.md` § Notation addendum — K = slots,  $\lambda$  = weight on L\_C, default K/N = 10%.

Not this file: Agent status in `model_OPORD.md`, doc 03, COMPASS stubs.

## Locked — already on disk (do not redo unless asked)

| Artifact                  | Location                                                                                | Notes                                                                 |
|---------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Phase B hand deck         | <code>HEROs_and_PASSes/slides/CHAR_Phase_B_characterization.pptx</code>                 | Scripts never overwrite — you format in PowerPoint                    |
| Auto compare deck         | <code>HEROs_and_PASSes/slides/auto/CHAR_intro_characterization_AUTO.pptx</code>         | <code>./scripts/build_characterization_slides.sh --auto-slides</code> |
| Slide walkthrough         | <code>07_Phase_B_Characterization_Slides_Explained.md</code>                            | Seed 42, T_j, hard vs smooth L_C, sort-and-chop                       |
| PD16 takeaways            | <code>08_PD16_Alex_meeting_takeaways.md</code>                                          | Whiteboard sketches, calibration roadmap                              |
| $\lambda_{crit}$ memo     | <code>06_Lambda_threshold_and_KN_memo.md</code>                                         | Sort-and-chop; $\lambda_{crit} \approx 4/\gamma$                      |
| $\theta \times K/N$ sweep | <code>HEROs_and_PASSes/theta/THETA_KN_sweep_*</code>                                    | Run with preset $\theta$ — revisit after $\theta(K/N)$                |
| $\gamma$ readable plots   | <code>HEROs_and_PASSes/sort_chop_lambda/GAMMA_sweep_lambda_curves_key_arms.png</code>   | Full grid: <code>GAMMA_sweep_lambda_curves.png</code>                 |
| Pass A/B/C gallery PNGs   | <code>HEROs_and_PASSes/pass_a/</code> , <code>pass_b/</code> , <code>pass_c_rho/</code> | Refresh: <code>./scripts/rebuild_hero_gallery.sh</code>               |
| Regenerate char PNGs      | <code>./scripts/build_characterization_slides.sh</code>                                 | Then Change Picture in hand deck if plots changed                     |

## Phase B — Characterization deck (built — your sign-off)

Agent/build status: scripts and PNGs exist for all rows below except A\_i and t\_j robustness. Your job: approve each claim in the hand deck and mark `[x]` when you can defend it to Alex.

| Done | Item                      | What “done” means                                                               | Proof                                                               |
|------|---------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------|
| [ ]  | Intro / glossary          | You can walk equations, benchmarks (A_i, T_j), seed 42, “not curve-fitting”     | Hand deck slide 0; doc 07 SLIDE 0                                   |
| [ ]  | $\rho$ OAT                | Both curves move; “assortativity matters”                                       | Hand deck slides 1–2; <code>pass_c_rho/</code>                      |
| [ ]  | $\lambda$ OAT             | Talent-only vs $\lambda > 0$ bend on A_i-binned curve                           | Hand deck slide 3; <code>pass_b/PASS_B_lambda_ablation_*.png</code> |
| [ ]  | $\theta$ OAT              | Sigmoid center sweep (still preset $\theta=0.72$ in deck)                       | Hand deck slides 4–5                                                |
| [x]  | $\theta \times K/N$ panel | Co-movement before fixing $\theta$ rule                                         | <code>theta/THETA_KN_sweep_*</code>                                 |
| [ ]  | $\gamma + \lambda_{crit}$ | Sort-and-chop; readable $\gamma$ sweep; $\lambda_{crit} \approx 4/\gamma$ story | Hand deck slides 6–7; doc 07                                        |
| [x]  | Walkthrough PDF           | Plain-English deck guide for you / Alex                                         | doc 07 (+ your PDF if printed)                                      |
| [x]  | PD16 briefing             | Alex saw deck; direction captured                                               | Aug 4 meeting; doc 08                                               |
| [ ]  | A_i distribution          | Robustness: default + 1–2 alternatives (SI note)                                | Not built                                                           |
| [ ]  | t_j distribution          | Empirical ballpark + brief sensitivity                                          | Not built                                                           |

Phase B “characterized” when: you mark the rows above [x] (except A\_i/t\_j if deferred) and Alex agrees the OAT deck answered “what moves when I turn this knob?” — not “we fit to data.”

PD16 follow-ups — Alex’s new direction (priority order)

Implement via shell env vars in gallery\_knobs.py ( GALLERY\_LC\_MODE , GALLERY\_THETA\_MODE ) — not a new conda environment. Defaults unchanged = current deck PNGs.

| Done | Priority                      | What to do                                                                                                                    | Proof                             |
|------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| [ ]  | 1. Team-level L_C             | Congestion = property of team j (same for all roster players); compare Pass B curves to LOO default                           | code path + PNG diff              |
| [ ]  | 2. $\theta$ from K/N          | Naive-draft quantile: $\theta = F_{A^{-1}}(1 - K/N)$ on ability draw (Beta in sim); re-run $\theta$ OAT + $\theta \times K/N$ | helper + updated PNGs             |
| [ ]  | 3. L_C distribution vs $\rho$ | Whiteboard Sketch A: histogram # teams vs L_C per $\rho$ arm; 2D heatmap team ability vs L_C                                  | new script + slide gap filled     |
| [ ]  | 4. $\rho$ low arm + intuition | Try $\rho \rightarrow 0.001$ ; one clean sentence why $\rho$ matters in this construction                                     | PNG + footer note on $\rho$ slide |
| [ ]  | 5. Update intro / doc 07      | When $\theta$ provenance shifts (539 preset $\rightarrow$ K/N quantile), refresh intro slide + walkthrough                    | slide + doc 07 § $\theta$         |
| [ ]  | 6. Env-var walkthrough        | You can export GALLERY_LC_MODE=... and regenerate without agent hand-holding                                                  | your terminal notes / date        |

PD16 implementation done when: team L\_C +  $\theta(K/N)$  modes run side-by-side with current deck; L\_C-vs- $\rho$  figures exist; you and Alex agree  $\rho/\theta$  are data-identifiable in principle.

Lingering from PD15 — still open (lower priority unless Alex asks)

Phase A gallery ( So\_Far\_.pptx slides 1–3) — separate from Phase B deck; never fully checked off.

| Done | Step                           | What to do                                                       | Proof      |
|------|--------------------------------|------------------------------------------------------------------|------------|
| [ ]  | $\lambda$ ablation panel       | Fix $\rho$ ; vary $\lambda$ (four values — mirror Pass C layout) | PNG path   |
| [ ]  | Sim double-plot                | One run: 16 bins on A_i (left) and poolq_loo (right)             | PNG path   |
| [ ]  | $\lambda$ overlay              | A_i-binned: $\lambda=0$ vs $\lambda>0$ overlay                   | PNG path   |
| [ ]  | Sort-and-chop on Pass C $\rho$ | Sort-and-chop arm on gallery $\rho$ figure                       | PNG path   |
| [ ]  | Fitted inputs on gallery       | Report t_j, $\sigma^2$ ; data-fitted vs sim-drawn                | slide note |
| [ ]  | Refresh So_Far_                | ./scripts/rebuild_hero_gallery.sh ; open in PowerPoint           | date       |

Optional checkpoint (pre-PD16):  $\rho \times \lambda$  interaction panel if a marginal sweep looked flat — ask Alex if still needed after team L\_C work.

Phase C — Calibration & fit (parked until PD16 rows 1–4 clear)

Alex PD16 roadmap: A\_i, K/N,  $\theta$ ,  $\rho$  from data; only  $\gamma$ ,  $\lambda$  remain free — fit via data properties, not “make sim curve match hero.”

| Done | Step                            | Pointer                                                       | Proof                |
|------|---------------------------------|---------------------------------------------------------------|----------------------|
| [ ]  | $\rho$ from data                | Match assignment assortativity in empirical data              | estimator + note     |
| [ ]  | $\theta$ from data              | Quantile at $1 - K/N$ on empirical or declared A distribution | ties to PD16 row 2   |
| [ ]  | $\gamma$ , $\lambda$ estimators | Principled fit — not hero-curve matching as definition        | memo / Alex sign-off |
| [ ]  | PD14 magnitude                  | 05_Alex_Magnitude_Spec.md — predictive importance A vs B      |                      |
| [ ]  | Notation cleanup                | $\sigma$ used twice (team $\sigma_j$ vs sigmoid)              | Model.pptx           |

Parked (good ideas, wrong moment)

| Item                             | Pointer                                     |
|----------------------------------|---------------------------------------------|
| Menger multiplicative congestion | Alex OK to park; 4-sentence memo if he asks |
| Preferential attachment sweep    | After boolean default validated             |
| Old 14-doc reading stack         | PARKED_FOR_LATER.md                         |
| Rebuild empirical hero from 530  | Not a gate                                  |
| Army / tenure figures            | Out of scope                                |

Explicitly not on this checklist

- Bin-for-bin LOO match as a gate
- B vs D decomposition (use L\_C only for now)
- Agent “Done” tables in OPORD / doc 03

Next action right now

1. Sign off Phase B deck — open CHAR\_Phase\_B\_characterization.pptx ; mark rows [x] as you approve each slide’s claim.
2. Start PD16 row 1 — team-level L\_C in code ( GALLERY\_LC\_MODE=team\_smooth when ready).
3. Read doc 08 if you need the whiteboard / calibration story in one sitting: 08\_PD16\_Alex\_meeting\_takeaways.md .
4. Defer Phase A gallery unless Alex asks for So\_Far\_.pptx refresh before team L\_C lands.